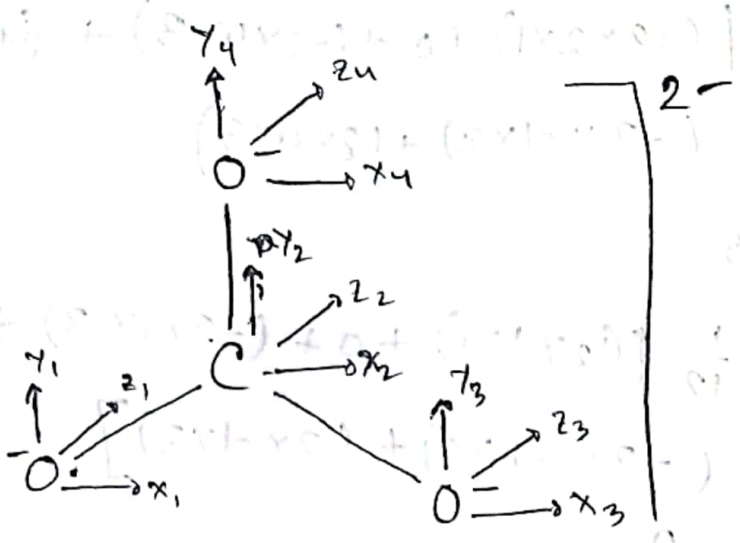
Chapter : 7



Using the x, y, and z direction on each atom as a basis. so the bassis of the representation is the set of 12 arrows.

Thus, the complete set of characters of reducible representation is-

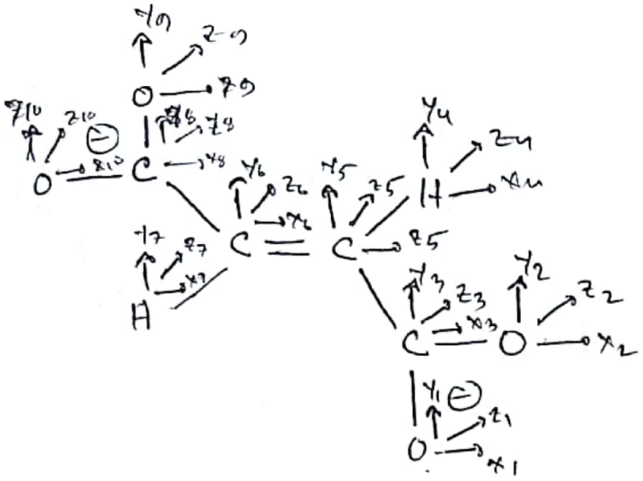
D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
	12	0	-2	4	-2	2

$$\text{No of } A_1' = \frac{1}{12}[(12\times1\times1) + 0 + (-2\times1\times3) + (4\times1\times1) + (-2\times1\times2) + (2\times1\times3)]$$

$$= 1$$

$$\text{No of } A_2' = \frac{1}{12}[(12\times1\times1) + 0 + (-2\times-1\times3) + (4\times1\times1) + (-2\times1\times2) + (2\times-1\times3)]$$

$$= 1$$



using x, y and z direction in each atom as basis. so the basis of the representation is a set of 30 arrows

The complete set of characters of reducible representation is

C_{2h}	E	C_2	i	σ_h
	30	0	0	10

No. of $A_g = \frac{1}{4}[(30\times1\times1) + 0 + 0 + (10\times1\times1)$

$= 10$

No of $B_g = \frac{1}{4}[(30\times1\times1) + 0 + 0 + (10\times1\times-1)$

$= 5$

No of $A_u = \frac{1}{4}[(30\times1\times1) + 0 + 0 + (10\times-1\times1)$

$= 5$

No of $B_u = \frac{1}{4}[(30 \times 1 \times 1) + 0 + 0 + 0 + 10]$

$= 10$

$\Gamma = 10A_g + 5B_g + 5A_u + 10B_u$

Symmetries of all molecular motion : $10A_g + 5B_g + 5A_u + 10B_u$

Symmetries of translations : $A_u + 2B_u$

Symmetries of Rotational motions : $A_g + 2B_g$

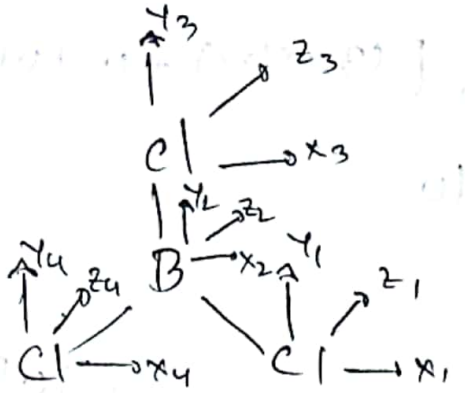
Symmetries of vibrations : $9A_g + 3B_g + 4A_u + 8B_u$

Raman active vibrations : $9A_g + 3B_g$

IR active vibrations : $4A_u + 8B_u$

* BCl_3





D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
	12	0	−2	4	−2	2

$$\Gamma = A_1' + A_2' + 3E' + 2A_2'' + E''$$

$$2E' + A_2''$$

Trans. →

$$2E''$$

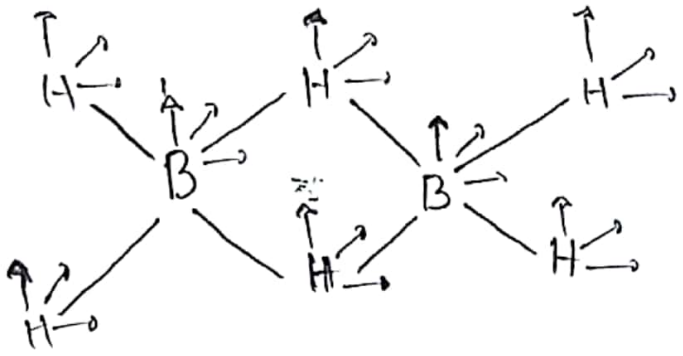
Rot → $A_2' +$

Vib → $A_1' + 2E' + A_2''$

Raman active → $A_1' + 2E'$

IR active → $2E' + A_2''$

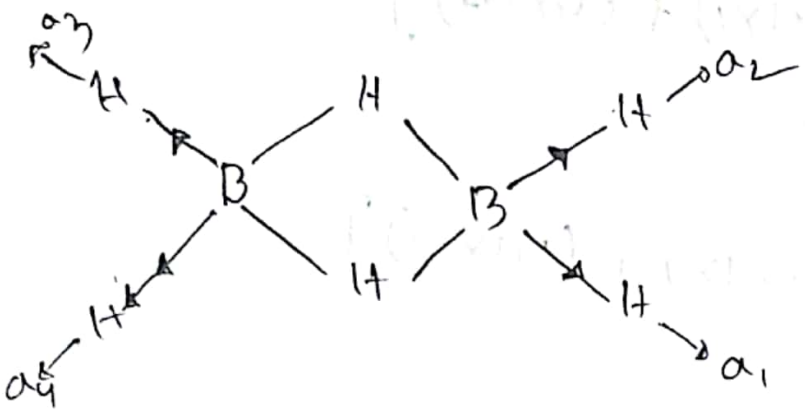
3.





D_{2h}	E	$C_2(z)$	$C_2(y)$	$C_2(x)$	i	$\sigma(xy)$	$\sigma(xz)$	$\sigma(yz)$
	24	-2						

using



Using, a_1, a_2, a_3, a_4 as a basis. The basis of the representation is a set of 4 arrows.

The complete set characteres of the reducible representation is

D_{2h}	E	$C_2(z)$	$C_2(y)$	$C_2(x)$	i	$\sigma(xy)$	$\sigma(xz)$	$\sigma(yz)$
	4	0	0	0	0	4	0	0

$$A_{1g} = \frac{1}{8}[(4\times1\times1) + 0 + 0 + 0 + 0 + (4\times1\times1) + 0 + 0]$$

$$= 1$$

~~$A_{2g} = \frac{1}{8}[(4\times1\times1) + (4\times1\times1)]$~~

$$= 1$$

$$B_{1g} = \frac{1}{8}[(4\times1\times1) + (4\times1\times1)]$$

$$= 1$$

$$B_{2g} = \frac{1}{8}[(4\times1\times1) + (4\times1\times-1)]$$

$$= 0$$

$$B_{3g} = \frac{1}{8}[(4\times1\times1) + (4\times1\times-1)]$$

$$= 0$$

$$A_u = \frac{1}{8}[4 - 4]$$

$$= 0$$

$$B_{1u} = 0$$

$$B_{2u} = 1$$

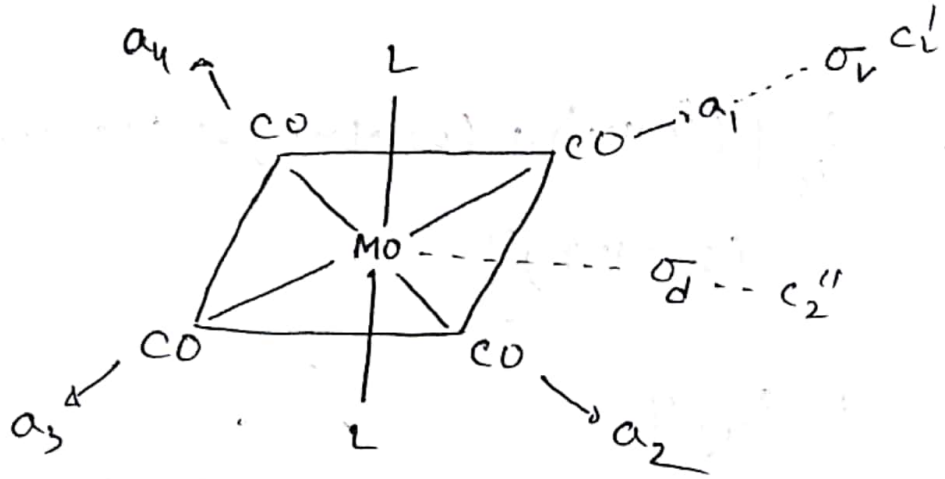
$$B_{3u} = 1$$

$\Gamma = 2A_g + B_{1g} + B_{2u} + B_{3u}$ = total modes of vib. (cause 'basis' along the bond)

~~Trans.~~ $B_{2u} + B_{3u}$

~~Rot.~~ ~~B_{1g}~~

~~Vib~~ ~~A_g~~



The four arrow a_1, a_2, a_3, a_4 shown, is the used as basis to represent the carbonyl streatching vibrations.

The complete set charatery of the reducible representation.

D_{4h}	E	$2C_4$	$C_2(=C_4^2)$	$2C_2'$	$2C_2''$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$
Γ	4	0	0	2	0	0	0	4	2	0

No of $A_{1g} = \frac{1}{16}[(4\times1\times1)+0+0+(2\times1\times2)+0+0+0+$
$(4\times1\times1)+(2\times1\times2)+0]$

$= 1$

No of $A_{2g} = \frac{1}{16}[(4\times1\times1)+0+0+(2\times1\times2)+0+0+0+$
$(4\times1\times1)+(2\times2\times-1)]$

$= 0$

No of. $B_{1g} = \frac{1}{16}[(4\times1\times1) + (2\times1\times2) + (4\times1\times1) + (2\times1\times2)]$

$= 1$

No of. $B_{2g} = \frac{1}{16}[(4\times1\times1) + (2\times-1\times2) + (4\times1\times1) + (2\times-1\times2)]$

$= 0$

No. of. $E_g = \frac{1}{16}[(4\times2\times1) + (4\times-2\times1)]$

$= 0$

No. of $A_{1u} = \frac{1}{16}[(4\times1\times1) + (2\times1\times2) + (4\times1\times1) + (2\times-1\times2)]$

$= 0$

No of $A_{2u} = \frac{1}{16}[(4\times1\times1) + (2\times-1\times2) + (4\times-1\times1) + (2\times1\times2)]$

$= 0$

No. of $B_{1u} = \frac{1}{16}[(4\times1\times1) + (2\times1\times2) + (4\times-1\times1) + (2\times-1\times2)]$

$= 0$

$$\text{No. of } B_{2u} = \frac{1}{16}[(4\times1\times1) + (2\times-1\times2) + (4\times-1\times1) (2\times1\times2)]$$

$$= 0$$

$$\text{No. of } E_u = \frac{1}{16}[(4\times2\times1) + (4\times2\times1)]$$

$$= 2$$

The irreducible representation

$$\Gamma = A_{1g} + B_{1g} + E_u$$

Condition to be ir active: It belongs to the same symmetry species as a component of dipole moment, i.e. to the same species as either x, y or z.

IR active vibrations: E_u

Condition to be Raman active: It belongs to the same symmetry species as a component of polarisability. To one of the binary

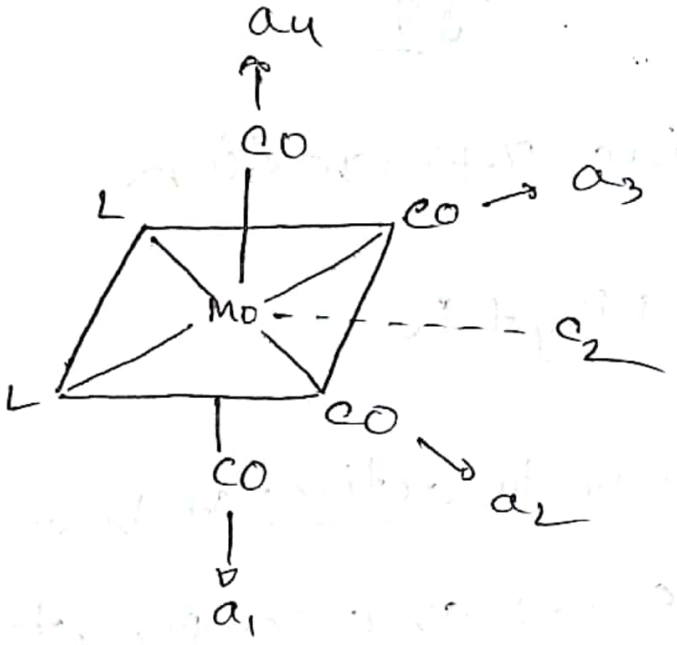
Products, x^2, y^2, z^2, xy, yz, xz or to a combination of products such x^2-y^2

Raman active vibrations: $A_{1g} + B_{1g}$

1 IR band (E_u)

2 Raman bands (A_{1g} and B_{1g})

*



The 4 arrows shown a_1, a_2, a_3, a_4 is used as basis.

The complete set of characters of the reducible rep

C_{2V}	E	C_2	σ	σ'
	4	0	2	2

No of $A_1 = \frac{1}{4}[(4\times1\times1) + 0 + (2\times1\times1) + (2\times1\times1)$

$= 2$

$A_2 = \frac{1}{4}[(4\times1\times1) + 0 + (2\times-1\times1) + (2\times-1\times1)]$

$= 0$

$B_1 = \frac{1}{4}[(4\times1\times1) + 0 + (2\times1\times1) + (2\times-1\times1)$

$= 1$

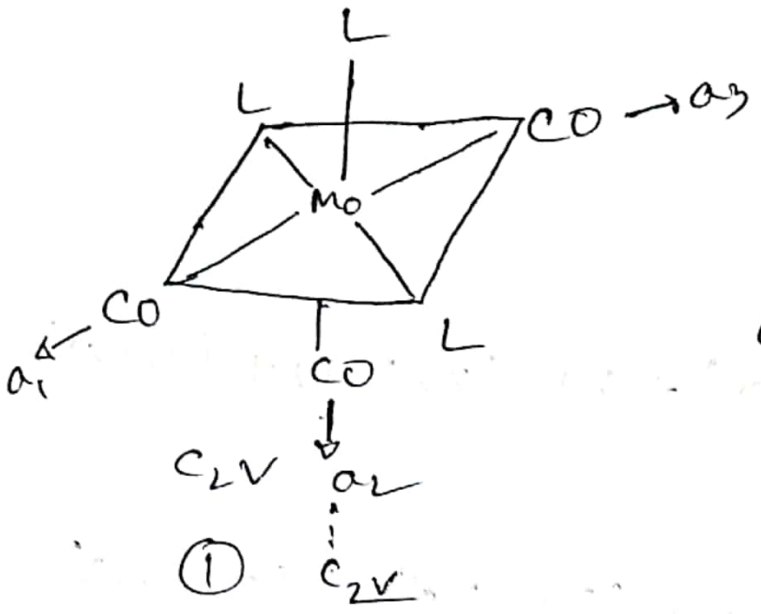
$B_2 = \frac{1}{4}[(4\times1\times1) + 0 + (2\times-1\times1) + (2\times1\times1)]$

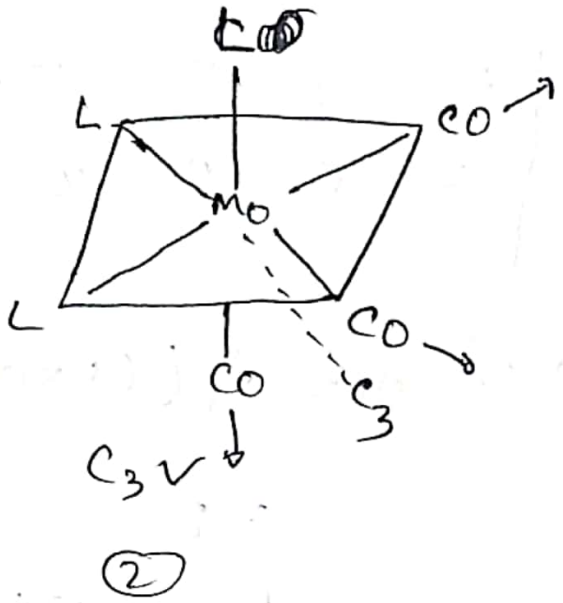
$= 1$

$\Gamma = 2A_1 + B_1 + B_2$

IR Bands: $2A_1 + B_1 + B_2$

Raman Bands: $2A_1 + B_1 + B_2$





For, ②

Using a_1, a_2, a_3 as basis.

The complete set of characters of reducible representation.

C_{3V}	E	$2C_3$	$3\sigma_v$
	3	0	1

$$A_1 = \frac{1}{6}\left[(3\times1\times1) + 0 + (1\times1\times3)\right] = 1$$

$$A_2 = \frac{1}{6}\left[(3\times1\times1) + 0 + (3\times1\times3)\right] = 0$$

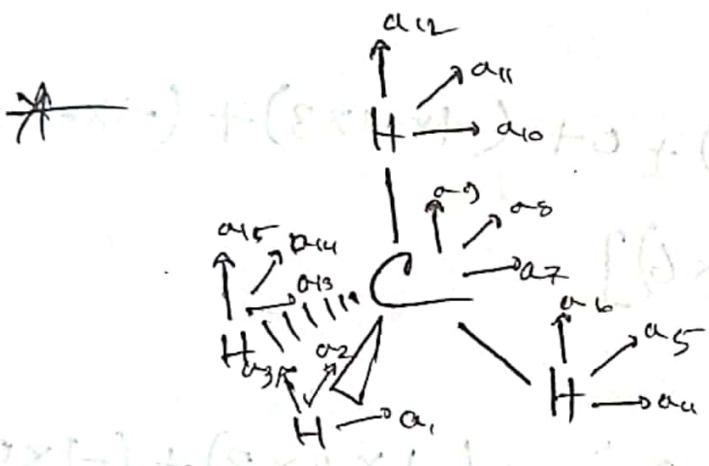
$$E = \frac{1}{6}\left[(3\times2\times1) + 0 + (3\times0\times1)\right]$$

$$2\ 1$$

$$\Gamma = A_1 + E$$

C_{3v}: 2 IR bands
2 Raman bands
} Coincident, $A_1 + E$

C_3



using x, y and z direction in each atom as a basis. The basis of the representation is a set of 15 arrows.

The complete set of characters of reducible representation

T_d	E	$8C_3$	$3C_2$	$6S_4$	$6\sigma_d$
	15	0	−1	−1	3

$$A_1 = \frac{1}{24}[(15\times1\times1) + (2\times0\times8) + (-1\times1\times3) + (-1\times1\times6) + (3\times1\times6)]$$

$$= A_1$$

$$A_2 = \frac{1}{24}[(15\times1\times1) + 0 + (-1\times1\times3) + (-1\times-1\times6) + (3\times-1\times6)]$$

$$= 0$$

$$E = \frac{1}{24}[(15\times2\times1) + 0 + (-1\times2\times3) + (-1\times0\times6) + 0$$

$$= E$$

$$T_1 = \frac{1}{24}[(15\times3\times1) + 0 + (-1\times-1\times3) + (-1\times1\times6) + (3\times-1\times6)]$$

$$= T_1$$

$$T_2 = \frac{1}{24}\left[(15\times3\times1) + 0 + (-1\times-1\times3) + (-1\times-1\times6) + (3\times1\times6)\right]$$

$$= 3$$

$$\Gamma = A_1 + E + T_1 + 3T_2$$

Trans = T_2

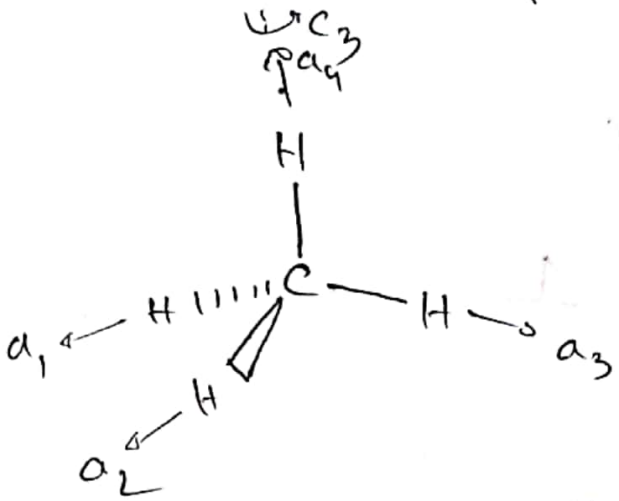
Rot = T_1

Vib: $A_1 + E + 2T_2$

IR: $2T_2$

Raman: $A_1 + E + 2T_2$

* CH_4 molecule is a tetrahedral molecule. It belongs to t_d point group.



Using (a_1, a_2, a_3, a_4) direction as basis, the character set of reducible representation

T_d	E	$8C_3$	$3C_2$	$6S_4$	$6\sigma_d$
Γ	4	1	0	0	2

$$\text{No. of } A_1 = \frac{1}{24}[(4\times1\times1) + (1\times1\times8) + 0 + 0 + (2\times1\times6)]$$

$$= 1$$

$$\text{No. of } A_2 = \frac{1}{24}[(4\times1\times1) + (1\times1\times6) + 0 + 0 + (2\times1$$

$$= 0$$

No. of $E = \frac{1}{24}[(1 \times 2 \times 1) + 3((1 \times (-1) \times 8) + 0 + 0$

$= 0$

No. of $T_1 = 0$

No. of $T_2 = 1$

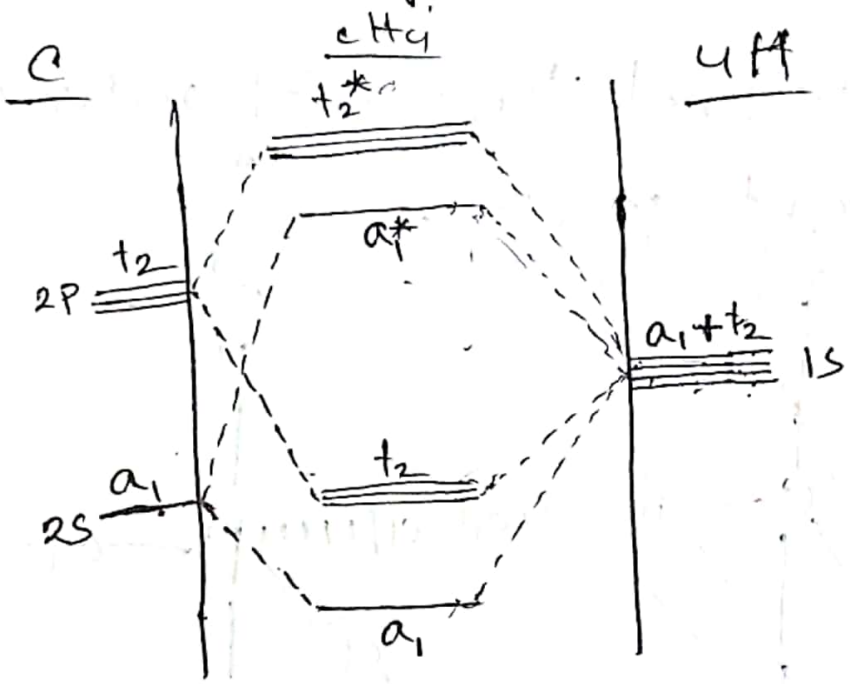
$\Gamma = A_1 + T_2$

From the character table we can see that

A_1 belongs to s orbital

T_2 belongs to P_x, P_y, P_z orbitals

correlation diagram



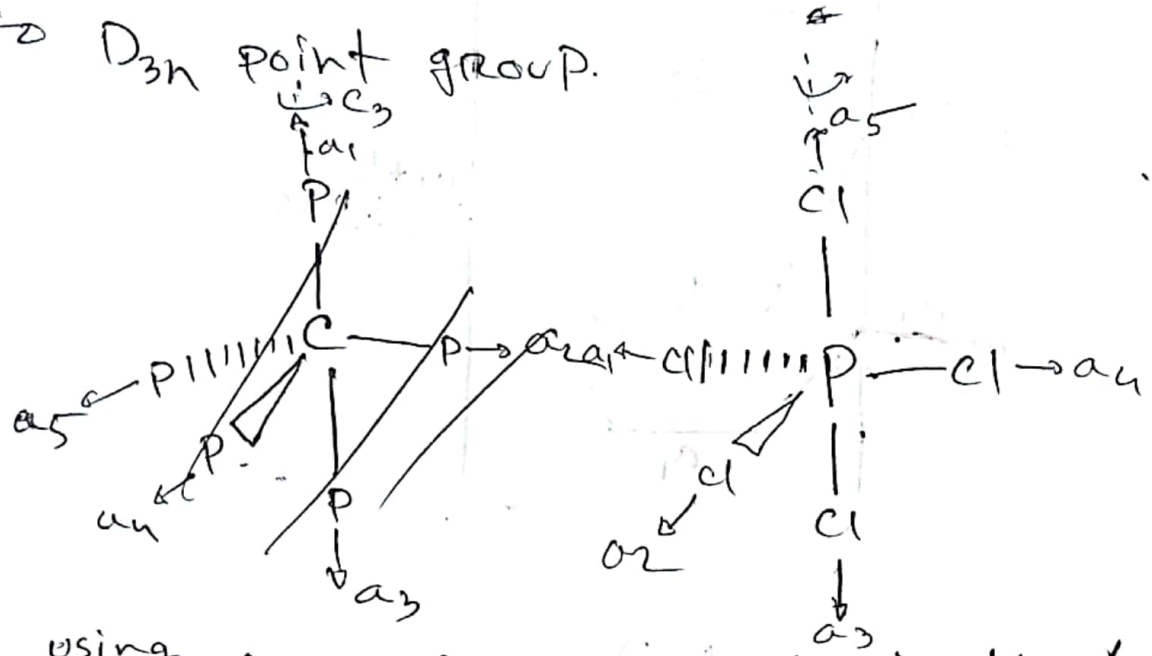
t_2^*, a_1^* } anti bonding

t_2, a_1 } bonding

C has four valance electrons in its outer shell and 4 H have four electron. we put (4+4) = 8 electrons into the molecular orbital.

PCl_5

It is a trigonal bipyramidal molecule, belongs to D_{3h} point group.



using $(a_1 - a_5)$ arrows as basis the

Let 5 PCl bond as basis to obtain a reducible representation in order to find the symmetry of the group.

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
	5	2	1	3	0	3

$$A' = \frac{1}{12}\left[(5\times1\times1)+(2\times1\times2)+(1\times3\times1)+(3\times1\times1)+0+(3\times1\times3)\right] = 2$$

$$A_2' = \frac{1}{12}[(5\times1\times1)+(2\times1\times2)+(-1\times1\times3)+(3\times1\times1)$$
$$+0+(3\times3\times-1)]$$

$$= 0$$

$$E' = 1$$

$$A_1'' = 0$$

$$A_2'' = 01$$

$$E'' = 0$$

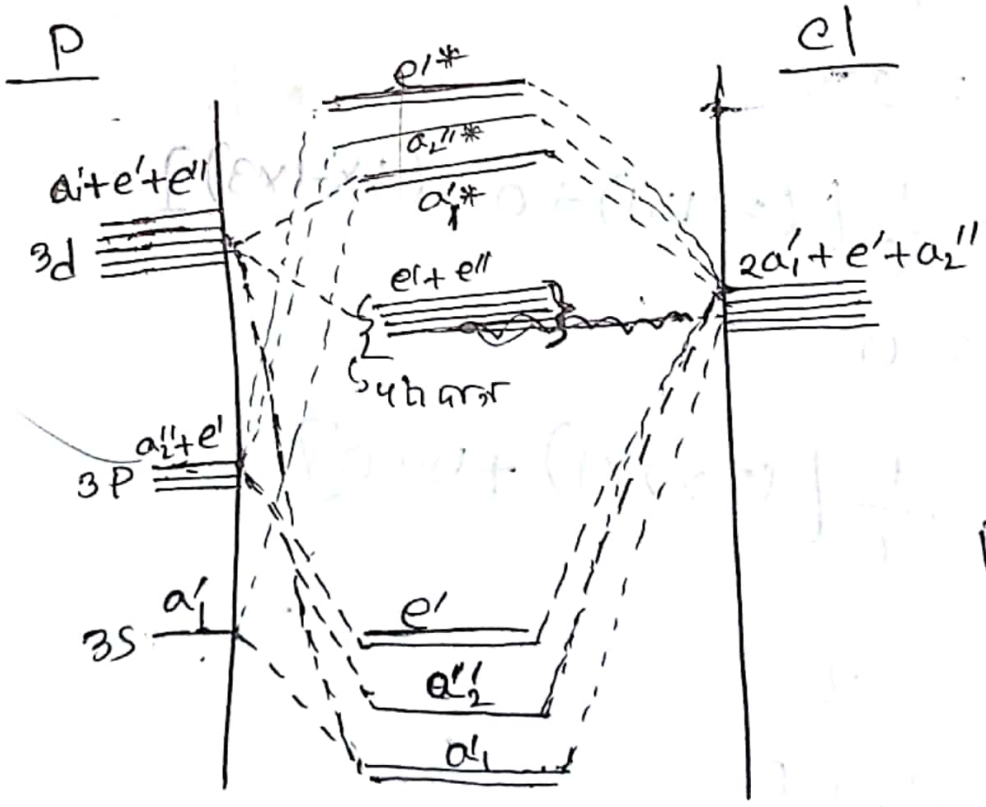
$s, d_{z^2} \longrightarrow A_1'$

$P_x, P_y \longrightarrow E' \longleftarrow d_{x^2-y^2}, d_{xy}$

$P_z \longrightarrow A_2''$

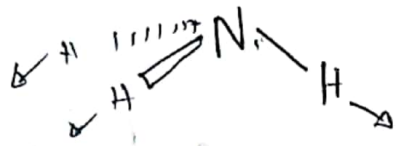
$d_{xz}, d_{yz} \longrightarrow E''$

$$\Gamma = 2A_1' + E' + A_2''$$



PCl_5

NH_3



Taking 3 bonds of NH_3 as basis.

C_{3v}	E	$2C_3$	$3\sigma_v$
	3	0	1

$$A_1 = \frac{1}{6}[(3\times1\times1) + 0 + (1\times1\times3)]$$

$$= 1$$

$$A_2 = \frac{1}{6}[(3\times1\times1) + 0 + (1\times-1\times3)]$$

$$= 0$$

$$E = \frac{1}{6}[(3\times2\times1) + 0 + 0]$$

$$= 2$$

$$\Gamma = A_1 + E$$

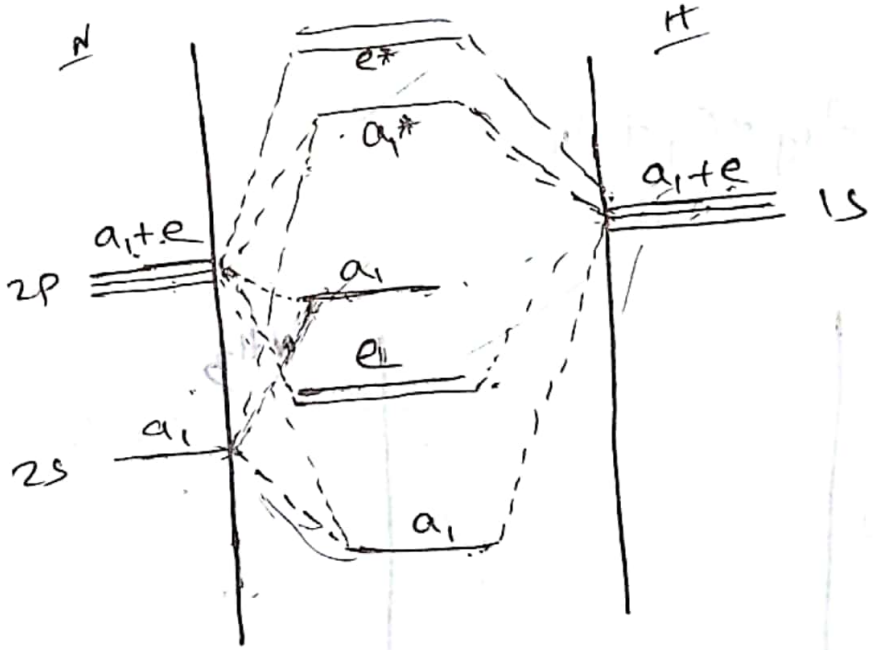
From the character table we see that

A_1 belongs to s, p_z, d_{z^2} orbital

E belongs to (x, y), p_x, p_y, , $d_{x^2-y^2}$, d_{xz}, d_{yz}, d_{xy}

since nitrogen has no energitically available d orbital, so the hybridisation of nitrogen for ammonia molecule is sp^3.

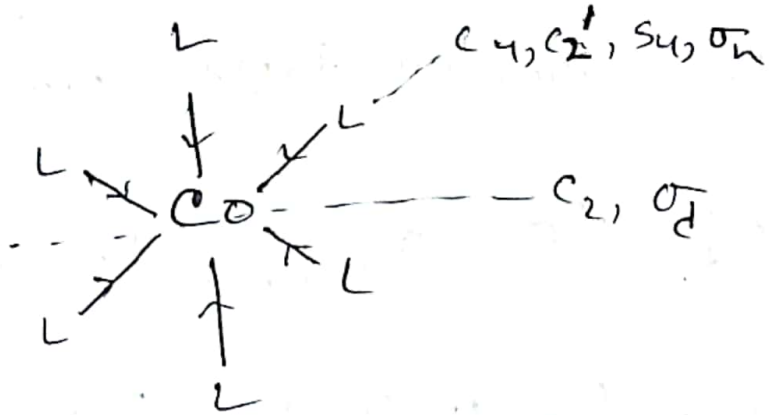
Correlation diagram



or (now)

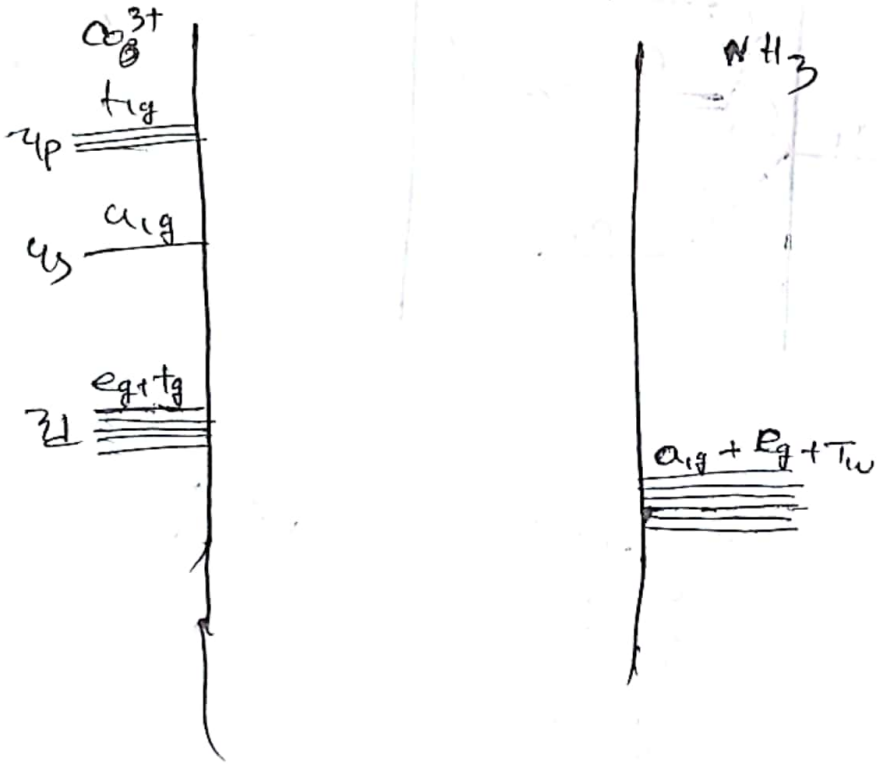
e

a_1

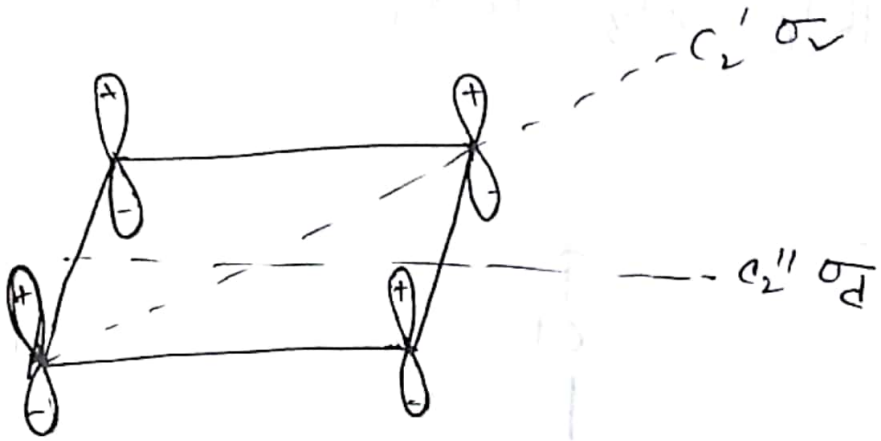


O_h	E	$8C_3$	$6C_2$	$6C_4$	$3C_2 (=C_4^2)$	i	$6S_4$	$8S_6$	$3\sigma_h$	$6\sigma_d$
	6	0	0	2	2		0	0	0	4	2

$\Gamma_7 = A_{1g} + E_g + T_{1u}$



2/



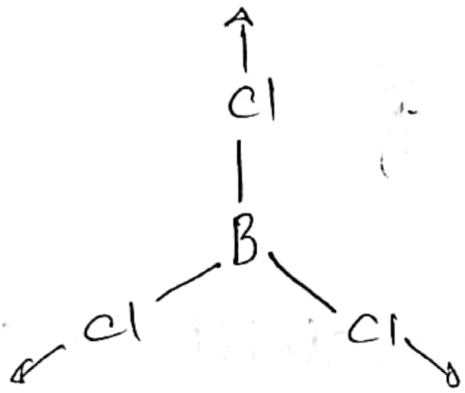
D_{4h}	E	$2C_4$	C_2	$2C_2'$	$2C_2''$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$
	4	0	0	-2	0	0	0	-4	2	0

$$\Gamma_5 = E_g + A_{2u} + B_{2u}$$

Suitable orbitals are:

H_2O, NH_3, PCl_5, BCl_3, CH_4 for chapter 6

BCl_3



The electronic configuration of central atom

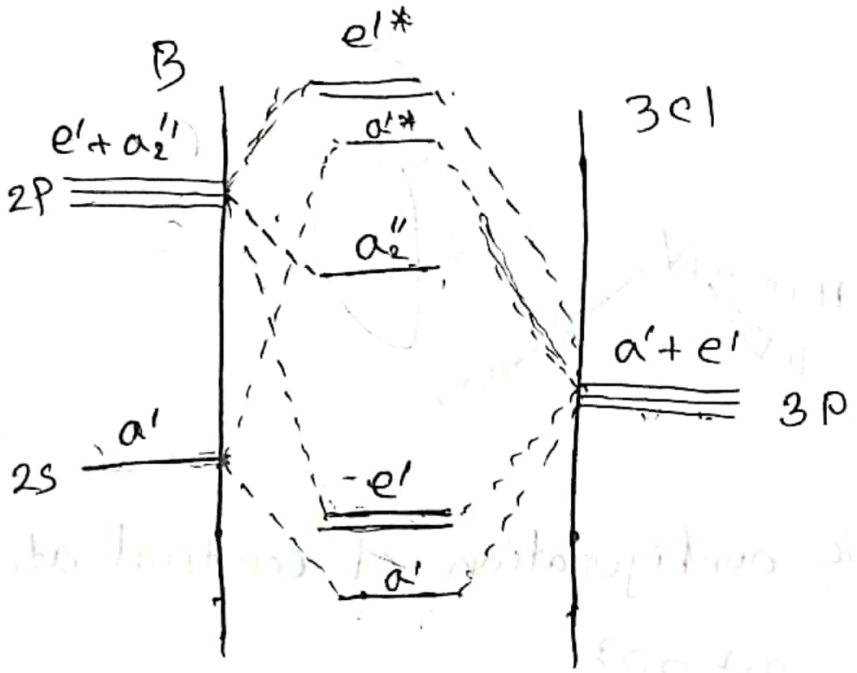
$B(5) = 1s^2\ 2s^2\ 2p^1$

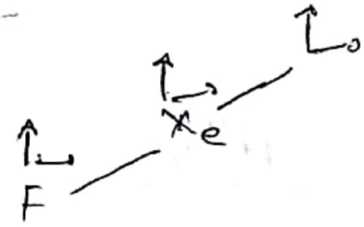
It belongs to D_{3h}

Let's take three vectors along the 3 B-Cl bond as basis to generate a reducible representation (Γ) for the D_{3h} (C_{2v}) Point group.

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
	3	0	1	3	0	1

$\Gamma = A' + E'$



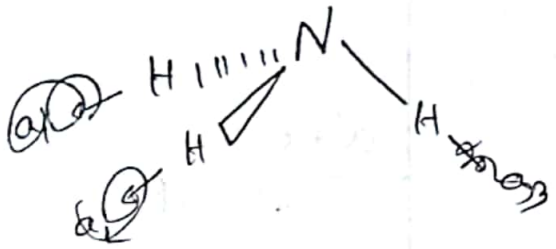


e'* } Anti bonding
a'* }

a2'' } — Non bonding

e' } — Bonding
a' }

NH_3

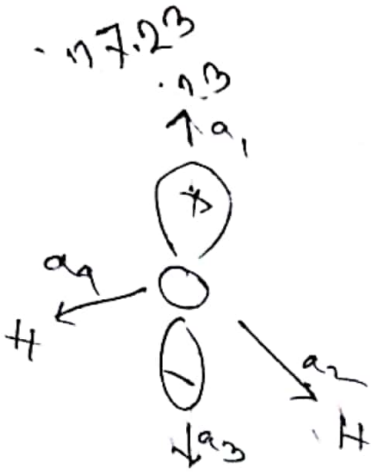


The electronic configuration of central atom

$N(7) = 1s^2\ 2s^2 2p^3$

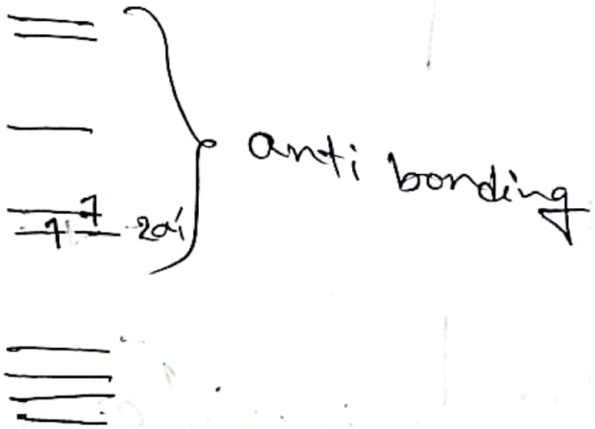
It belongs to C_{3v} point group

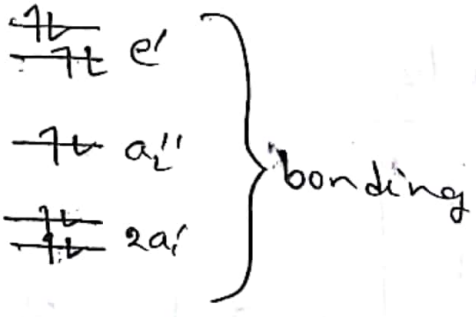
Let's take three vectors along the three N-H bonds as basis to generate a reducible representation (Γ) for C_{3v} point group



C_{2v}	E	C_2	σ	σ'
Γ	4	2	1	2

$$A_1 = \frac{1}{4}$$





Chapter 6

NH_3

C_{3v}	
	3 01 ~~30~~

H_2O

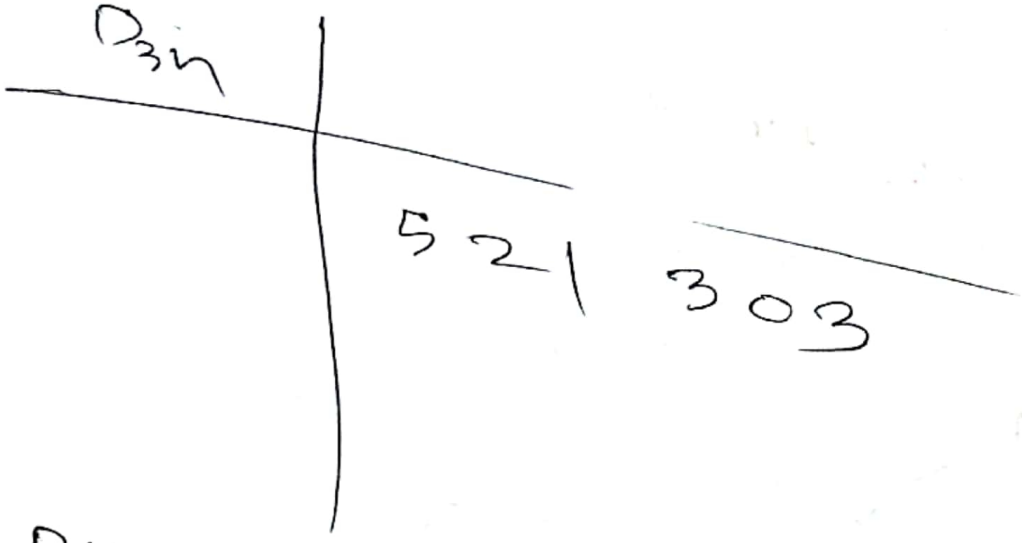
C_{2v}	
	20 20

H_2O	E C_2
	4 0 2

BCl_3

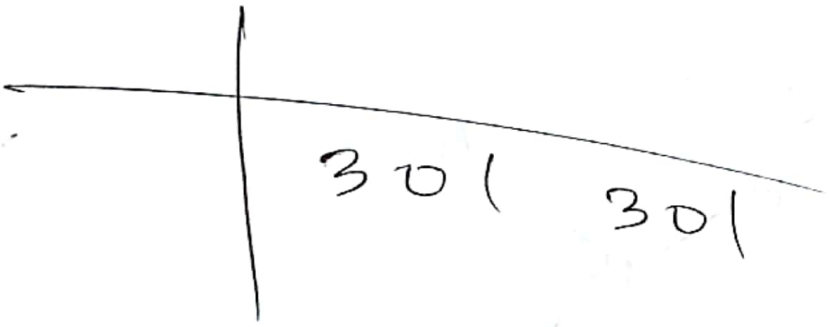
D_{3h}		
	30	30

PCl_5

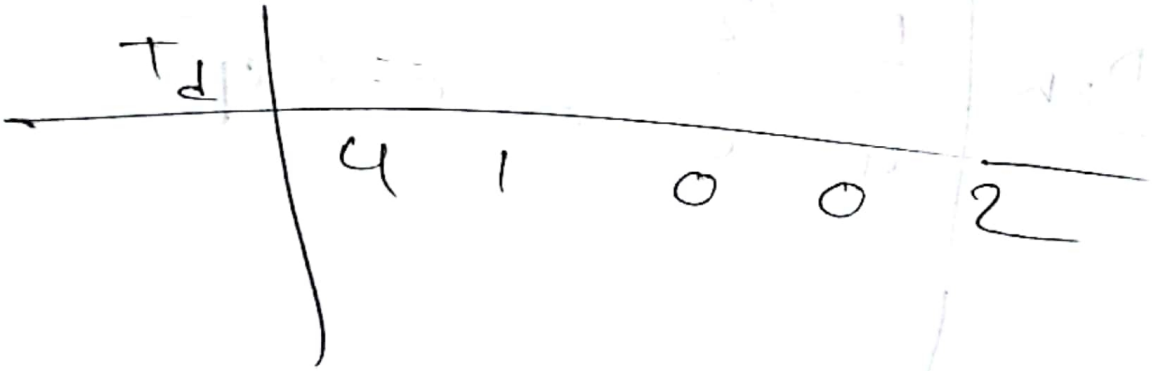


-0.7

BH_3

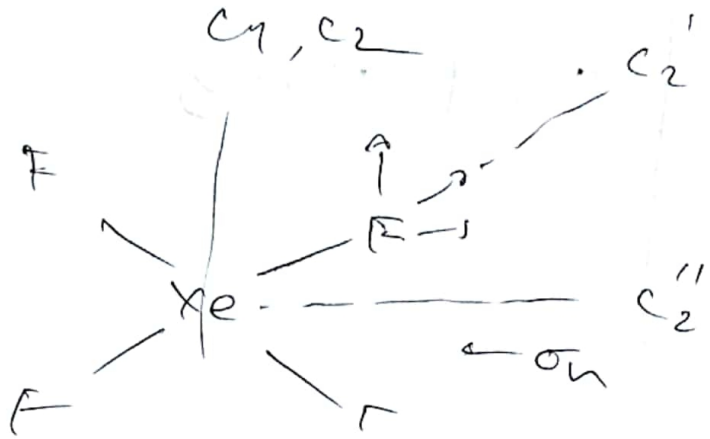


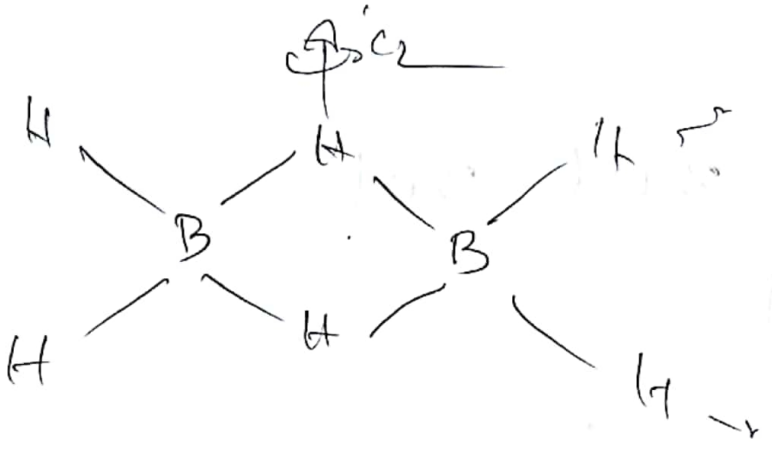
CH_4



chapter 7

XeF_4

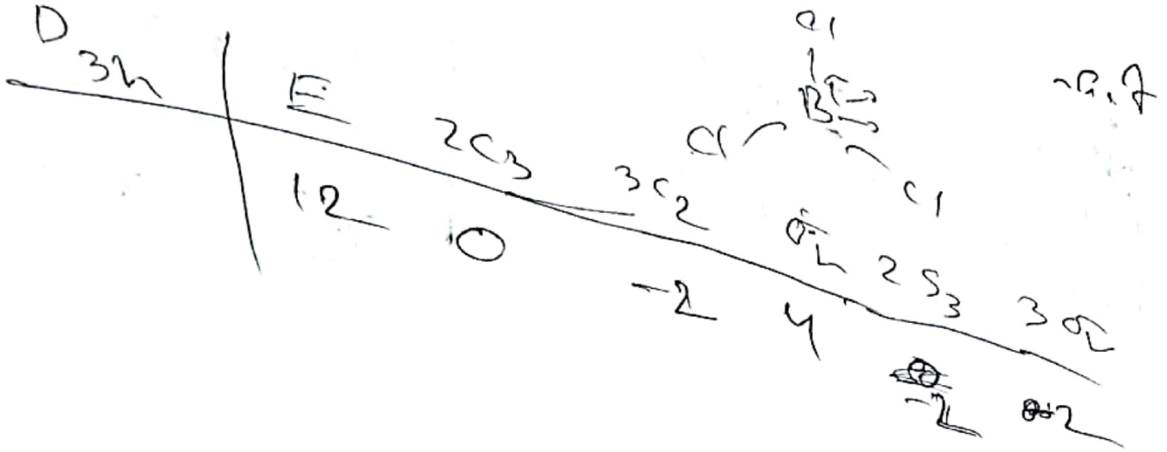


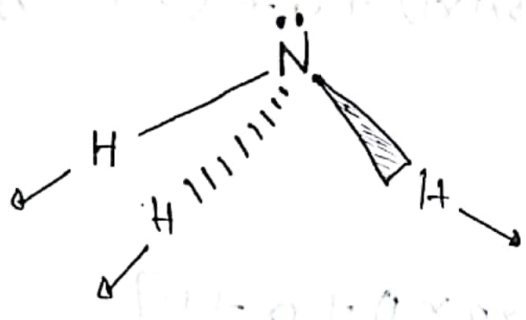


D_{2h}	E	C_2	C_2'	C_2''	i	σ_{xy}	σ_{xz}	σ
	4	0	0	0	0	4	0	

BCl_3 $-1+2\cos 120$

2.





Let us take three vectors along (N-H) bonds as basis to generate the reducible representation of the C_{3V} point group.

C_{3V}	E	C_3	σ
Γ	3	0	1

Let us take the character set of the reducible representation by using reduction formula.

No of $A_1 = \frac{1}{6}\left[(3\times1\times1) + 0 + (1\times1\times3)\right]$.

$= \frac{6}{6}$

$= 1$

No of $A_2 = \frac{1}{6}[(3\times1\times1) + 0 + (3\times1\times-1)]$

$= 0$

No of $E = \frac{1}{6}[(3\times2\times1) + 0 + 0]$

$= 1$

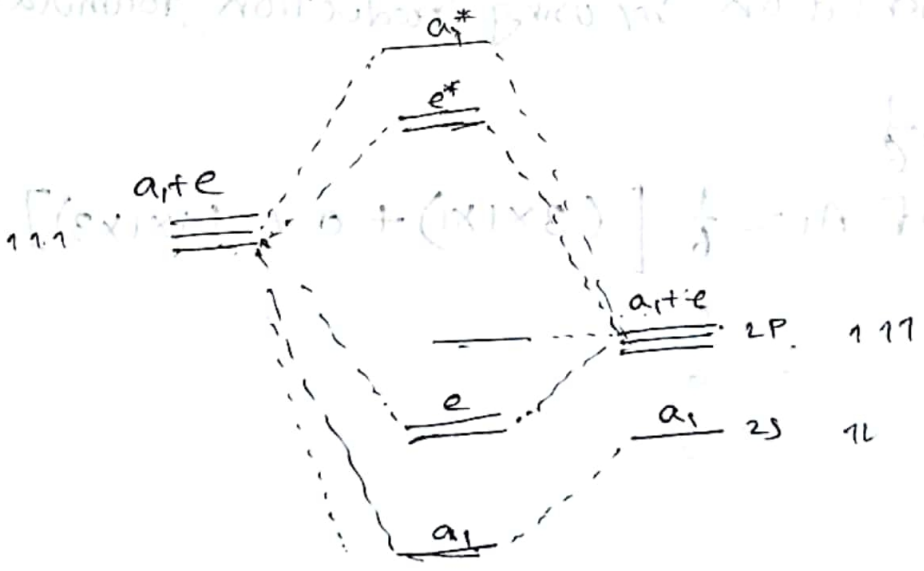
So, $\Gamma = A_1 + E$

From the character table,

A_1 belongs to P_z and S orbital

E belongs to P_x and P_y orbital

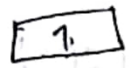
Correlation diagram of ammonia



N= $1s^2$ $2s^2$ $2p^3$

H= $1s^1$





——— a_{1g}^*

— — e_g^*

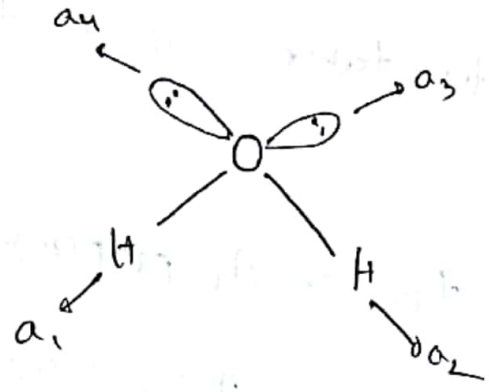
1↓ non

1↓ ~~1↓~~ e_g

1↓ a_{1g}

* $[M(CO)_3L_3]$ molecule is an octahedral molecule. It belongs to the

Chapter-6



→ Basis a_1, a_2, a_3, a_4

→ Point group C_{2v}

The complete set of characters of the reducible representation by using these basis

C_{2v}	E	C_2	σ	σ'
Γ	4	0	2	2

$$A_1 = \frac{8}{4} = 2$$

$$A_2 = 0$$

$$B_1 = 1$$

$$B_2 = 1$$

$\Gamma' = 2A_1 + B_1 + B_2$

From the character table the suitable orbitals is,

s, d_{z^2}, P_z belongs to A_1 representation

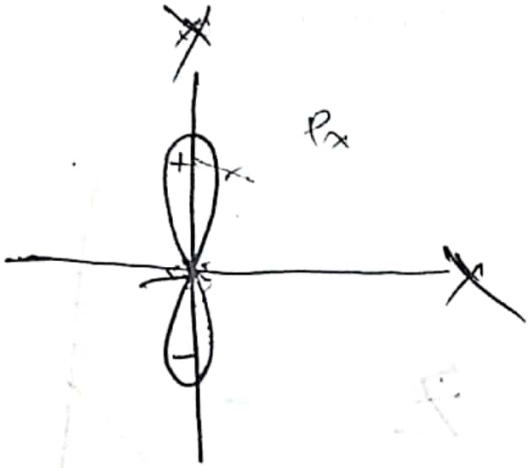
P_x, d_{xz} belongs to B_1 representation

P_y, d_{yz} belongs to B_2 representation.

Since there is no energetically available d orbital, so the hybridization of oxygen for water.

sp^3

P_x, P_y, P_z



For, P_x orbital

C_{2v}	E	C_2	i	σ_h	
B_u	1	−1	−1	1	x

1. −1 −1 1

